Chapter 3 (10 questions)

1. Write an Arduino sketch that reads an analog sensor value (0–1023) on pin A0 and uses the basic arithmetic operators (+, -, *, /) to convert it into a voltage value (0–5 V). Explain why integer division would give a wrong result here and how you corrected it.

Ans:

```
void setup() {
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(A0);          // 0 to 1023
  float voltage = sensorValue * (5.0 / 1023.0); // scale to 0–5 V
  Serial.print("Sensor: "); Serial.print(sensorValue);
  Serial.print("  Voltage: "); Serial.println(voltage);
  delay(500);
}
```

2. Two analog sensors provide readings through A0 and A1. Write an Arduino code fragment using abs() to calculate the difference between the readings and turn ON an LED if their difference is greater than 20.

Answer:

```
int sensor1 = analogRead(A0);
int sensor2 = analogRead(A1);
int difference = abs(sensor1 - sensor2);
if (difference > 20)
{
  digitalWrite(LED_BUILTIN, HIGH);
}
else
{
  digitalWrite(LED_BUILTIN, LOW);
}
```

3. Using the modulo operator (%), write a code fragment that blinks an LED only on every 5th iteration of a loop counter. Explain how the remainder operation makes this possible.

Ans: int counter = 0;const int ledPin = 13;

```
void setup() { pinMode(ledPin, OUTPUT); }
void loop() {
  counter++;
  if (counter % 5 == 0) {          // remainder is 0 every 5th time
    digitalWrite(ledPin, HIGH);
    delay(200);
    digitalWrite(ledPin, LOW);
  }
  delay(200);
}
```

4. Demonstrate the use of the increment (++) and decrement (--) operators. Write a short sketch that counts up from 0 to 10 and then back down to 0, printing each value to the Serial Monitor. Q5. Write a sketch that uses the map() function to scale a potentiometer reading (0–1023) to an LED brightness range (0–255) for PWM output. Explain each of the five arguments of map().

Ans:

```
void setup() { Serial.begin(9600); }

void loop() {
  for (int i = 0; i <= 10; i++) Serial.println(i);   // count up
  for (int i = 10; i >= 0; i--) Serial.println(i);   // count down
  delay(2000);
}
```

5. Explain and demonstrate the min() and max() functions. Write a code fragment that reads two sensor values and prints both the smaller and the larger of the two.

Ans:

```
void setup() { }
void loop() {
  int sensorValue = analogRead(A0);              // 0–1023
  int brightness = map(sensorValue, 0, 1023, 0, 255); // scale to 0–255
```

```
  analogWrite(9, brightness);                  // PWM to LED on pin 9
}
```

6. Develop an Arduino program that generates 10 random numbers between 0 and 9. Use randomSeed() to initialize the random-number sequence.

Answer:

```
void setup()
{
  Serial.begin(9600);
  randomSeed(analogRead(A0));
  for (int i = 0; i < 10; i++)
  {
    Serial.println(random(10));
  }
}
void loop()
{
}
```

7. Use one byte variable to represent the ON/OFF status of eight devices. Write an Arduino code fragment using bitSet(), bitClear(), and bitRead() to control and verify individual device states.

Answer:

```
byteflags = 0;
// Turn ON device 2 and device 5
bitSet(flags, 2);
bitSet(flags, 5);
// Check device 2
if (bitRead(flags, 2))
{
  Serial.println("Device 2 is ON");
}
// Turn OFF device 2
bitClear(flags, 2);
// Check device 2 again
if (bitRead(flags, 2) ==0)
```

```
{
  Serial.println("Device 2 is OFF");
}
```

8. An Arduino system needs to transmit the integer value 258 (0x0102) as two separate bytes. Write a program using highByte() and lowByte() to extract the bytes and then reconstruct the original value using word().

Answer:

```
intvalue = 258;       // 0x0102
bytehigh = highByte(value);
bytelow  = lowByte(value);
Serial.print("High byte = ");
Serial.println(high);
Serial.print("Low byte = ");
Serial.println(low);
// Reconstruct the original value
intreconstructed = word(high, low);
```

9. An Arduino program stores the value 6 in an integer variable.

```
intx = 6;
x = x<<1;
x = x>>2;
```

Write the binary representation of 6.

Determine the value of x after the left shift.

Determine the final value of x after the right shift by 2 positions.

Explain how left and right shifting are related to multiplication and division by powers of 2.

Answer:

6 = 0110

After x << 1:

0110 → 1100 = 12

After x >> 2:

1100 → 0011 = 3

Therefore, the final value of x = 3.

x << 1 is equivalent to x × 2

x >> 2 is equivalent to x ÷ 4

Serial.print("Reconstructed value = ");

Serial.println(reconstructed);

10. Design an Arduino code fragment that processes a counter from 1 to 20 and turns an LED ON only when the counter reaches a multiple of 5. Use the modulo (%) operator and explain how it identifies the required values.

Answer:

```
for (int counter = 1; counter <= 20; counter++)
{
  if (counter % 5 == 0)
  {
    digitalWrite(LED_BUILTIN, HIGH);
    Serial.println("LED ON");
  }
  else
  {
    digitalWrite(LED_BUILTIN, LOW);
  }
}
```